N83 Koblitz: index calculus against precomputed-table rho

Matched-precomputation control for the 12.29× (K600) and 113× (K3300) "IC beats rho" readings · 2026-10-16

1. Question and verdict

The N83 compact-orbit results compare IC's **online** interval, which starts after 600–800 s of reusable setup holding a multi-GiB state table, with Pollard rho whose distinguished-point table starts empty for every target. Here rho gets the same reusable precomputation (Bernstein–Lange): a table of `G`-only distinguished points built before any target is seen, on the repository's fastest rho step (fold backend), under the protocol in `PROTOCOL.md` (frozen before timing; amendment 1 before run 2). Same 16 public hash-to-curve points as hosted run `gha-38597929540-1`, same hosted runner for every arm within a job.

Verdict (decision rule of `PROTOCOL.md`). Neither online "IC beats rho" reading survives a matched-precomputation rho. For every time-matched and memory-matched comparison, the 95% interval of the median paired ratio IC/precomputed-rho lies entirely above 1: precomputed rho is faster online.

claim	matched arm (run, runner)	median IC / rho-with-table	95% CI	rule
K3300, 113×	(a′) = IC wall 680 s, 1 thread (run 2, EPYC 7763)	3.74	[2.65, 11.05]	does not survive
K3300, 113×	(b) = IC max RSS 10.79 GB (run 2, EPYC 7763)	40.61	[24.27, 71.72]	does not survive
K3300	(b) (run 1, EPYC 9V45)	42.72	[25.49, 74.91]	does not survive
K600, 12.29×	(a′) = IC wall 772 s (run 2, Xeon 6973P-C)	81.52	[46.35, 236.59]	does not survive
K600, 12.29×	(a) 650 s ≤ IC wall 785 s (run 1, EPYC 7763)	214.73	[47.94, 483.84]	does not survive
K600, 12.29×	(b) = IC max RSS 5.75 GB (run 2 / run 1)	395.53 / 324.88	[177.9, 1134.5] / [148.1, 936.2]	does not survive

Run 1′s k3300 arm (a) (650 s on an EPYC 9V45 where IC's setup took 498 s) is not time-matched and is excluded from the verdict (amendment 1); it reads 16.26 [7.23, 22.60]. Positive control: on the same runners the empty-table rho is slower than IC, 70.9× [19.2, 92.4] (K3300, run 2, EPYC 7763) and 3.62× [0.93, 4.48] (K600, run 2), so the jobs reproduce the direction of the original readings. The K600 reading is 3.6–5.3× here against 12.29× originally because this rho is the fold worker; the hosted run's reference was the slower strong rung 3.

Operation counts (AGENTS.md §5a). Wall time is the claims' metric and the verdict's. On counted group-addition equivalents no ranking is admitted: K600 IC emits probes and root calls, and the IC/rho order flips between 0 and 0.1 GAE per native operation (IC's counted additions are about 5 per target when its probes are free); K3300 IC (`koblitz_orbit_dlp_fast.rs`) emits probes but no root-call count, so its count is a lower bound and cannot be ranked. The bands are in `analysis.json` (`operation_count_band`).

2. Results

2.1. run 2 (38878565324) — IC K600 job (Intel(R) Xeon(R) 6973P-C)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: `true` (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	563.884	[165.222, 1277.173]	1966.983	[582.617, 3574.033]
rho, empty table (fold, Tier 1 worker)	1340.162	[927.073, 2401.124]	1620.925	[1182.901, 2092.605]
rho + table, (a) 650 s, seed …601	2.337	[1.406, 2.83]	2.406	[1.808, 3.057]
rho + table, (a′) = IC wall, seed …605	5.172	[3.381, 9.073]	5.91	[4.396, 7.622]
rho + table, (b) = IC max RSS	1.189	[0.887, 1.773]	1.729	[1.074, 2.548]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	296.64	[64.68, 657.79]	817.56	[232.1, 1619.86]	16/16
ic / bl_a2_wall_equal	81.52	[46.35, 236.59]	332.84	[89.97, 708.38]	16/16
ic / bl_b_memory_matched	395.53	[177.93, 1134.45]	1137.58	[309.35, 2697.39]	16/16

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
rho_plain_fold / bl_a_time_matched	786.74	[387.16, 1112.35]	673.72	[459.69, 932.67]	16/16
rho_plain_fold / ic	3.62	[0.93, 4.48]	0.82	[0.4, 2.82]	11/16

precomputation	wall s	CPU s	memory	notes
IC	773.7	804.03	5.75 GB max RSS	rank threads 1
rho + table, (a) 650 s, seed …601	650.8	651	0.438 GB table	27354500 entries, 5.37 M steps/s, stop: wall_seconds
rho + table, (a′) = IC wall, seed …605	772.7	773	0.512 GB table	31981512 entries, 5.29 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	1904.4	7558	5.67 GB table	354379300 entries, 11.97 M steps/s, stop: table_bytes

2.2. run 2 (38878565324) — IC K3300 job (AMD EPYC 7763 64-Core Processor)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: `true` (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	32.784	[24.2, 67.529]	49.256	[31.481, 70.282]
rho, empty table (fold, Tier 1 worker)	1967.327	[1326.671, 3570.145]	2375.319	[1739.184, 3058.48]
rho + table, (a) 650 s, seed …601	3.396	[2.032, 4.739]	4.205	[2.749, 6.052]
rho + table, (a′) = IC wall, seed …605	8.113	[6.261, 12.137]	8.913	[6.826, 11.042]
rho + table, (b) = IC max RSS	1.167	[0.497, 1.598]	1.183	[0.836, 1.568]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	11.33	[4.08, 18.8]	11.71	[6.7, 21.47]	15/16
ic / bl_a2_wall_equal	3.74	[2.65, 11.05]	5.53	[3.31, 9.03]	14/16
ic / bl_b_memory_matched	40.61	[24.27, 71.72]	41.63	[24.71, 71.95]	15/16
rho_plain_fold / bl_a_time_matched	803.85	[401.15, 1115.98]	564.83	[330.72, 918.76]	16/16
rho_plain_fold / ic	70.9	[19.17, 92.37]	48.22	[26.71, 87.87]	16/16

precomputation	wall s	CPU s	memory	notes
IC	681.6	825.02	10.79 GB max RSS	rank threads 4
rho + table, (a) 650 s, seed …601	650.6	651	0.304 GB table	19008681 entries, 3.73 M steps/s, stop: wall_seconds
rho + table, (a′) = IC wall, seed …605	680.2	680	0.317 GB table	19825177 entries, 3.72 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	5293	21036	10.519 GB table	657416242 entries, 8.07 M steps/s, stop: table_bytes

2.3. run 1 (38855595549) — IC K600 job (AMD EPYC 7763 64-Core Processor)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: `true` (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	580.276	[168.726, 1340.436]	2024.844	[607.602, 3675.26]
rho, empty table (fold, Tier 1 worker)	1969.815	[1324.54, 3572.084]	2375.182	[1738.386, 3056.715]
rho + table, (a) 650 s, seed …601	3.333	[2.025, 4.7]	4.17	[2.721, 6.003]
rho + table, (b) = IC max RSS	1.471	[1.111, 2.218]	2.148	[1.34, 3.159]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	214.73	[47.94, 483.84]	485.57	[131.91, 1072.5]	16/16
ic / bl_b_memory_matched	324.88	[148.1, 936.24]	942.57	[258.37, 2226.11]	16/16
rho_plain_fold / bl_a_time_matched	806.03	[399, 1148.52]	569.58	[333.1, 928.35]	16/16
rho_plain_fold / ic	5.25	[1.26, 6.32]	1.17	[0.58, 3.97]	12/16

precomputation	wall s	CPU s	memory	notes
IC	784.7	816.40	5.75 GB max RSS	rank threads 1
rho + table, (a) 650 s, seed …601	650.6	651	0.304 GB table	18987498 entries, 3.72 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	2818.3	11203	5.67 GB table	354371105 entries, 8.07 M steps/s, stop: table_bytes

2.4. run 1 (38855595549) — IC K3300 job (AMD EPYC 9V45 96-Core Processor)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: `true` (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	24.774	[18.425, 50.21]	37.278	[23.802, 53.249]
rho, empty table (fold, Tier 1 worker)	1361.144	[914.186, 2456.497]	1636.179	[1200.606, 2108.434]
rho + table, (a) 650 s, seed …601	2.321	[1.4, 2.797]	2.33	[1.76, 2.968]
rho + table, (b) = IC max RSS	0.841	[0.391, 1.155]	0.86	[0.609, 1.14]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	16.26	[7.23, 22.6]	16	[9.23, 26.77]	15/16
ic / bl_b_memory_matched	42.72	[25.49, 74.91]	43.34	[25.79, 75.05]	15/16
rho_plain_fold / bl_a_time_matched	805.56	[399.17, 1147.27]	702.26	[484.03, 963.21]	16/16
rho_plain_fold / ic	63.45	[17.66, 85.23]	43.89	[24.22, 80.21]	16/16

precomputation	wall s	CPU s	memory	notes
IC	497.6	603.71	10.79 GB max RSS	rank threads 4
rho + table, (a) 650 s, seed …601	650.6	651	0.455 GB table	28434341 entries, 5.58 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	3126.3	12429	10.519 GB table	657460555 entries, 13.67 M steps/s, stop: table_bytes

3. Graphs

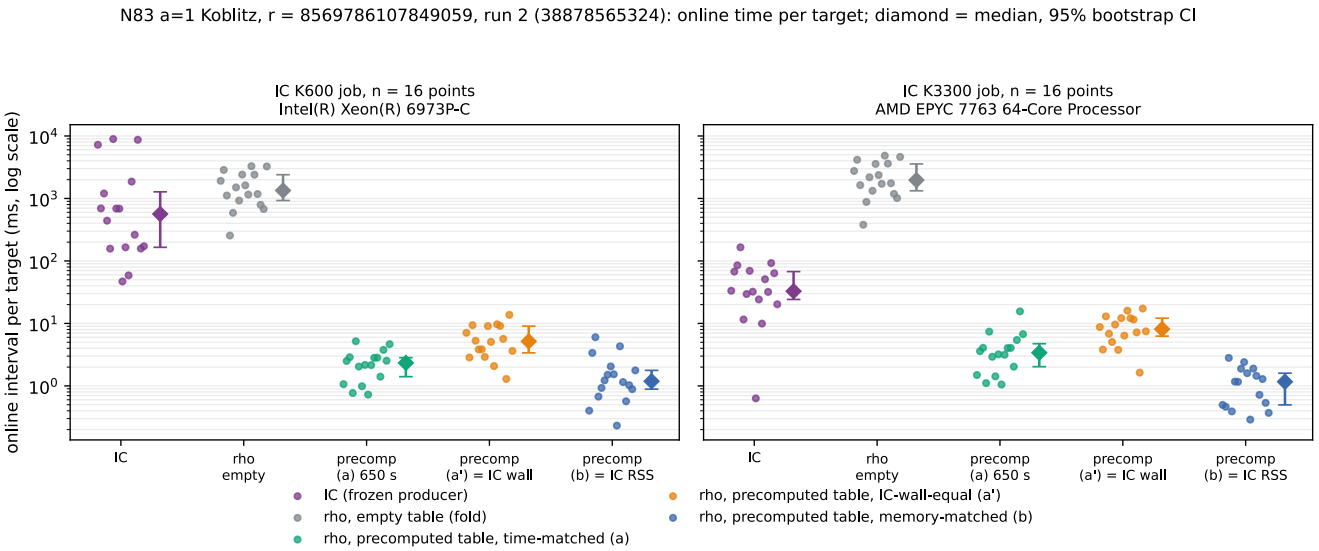


Figure 1: Run 2 (38878565324): online interval per target, every arm, both jobs. Diamonds: median with 95% bootstrap interval (10,000 target resamples).

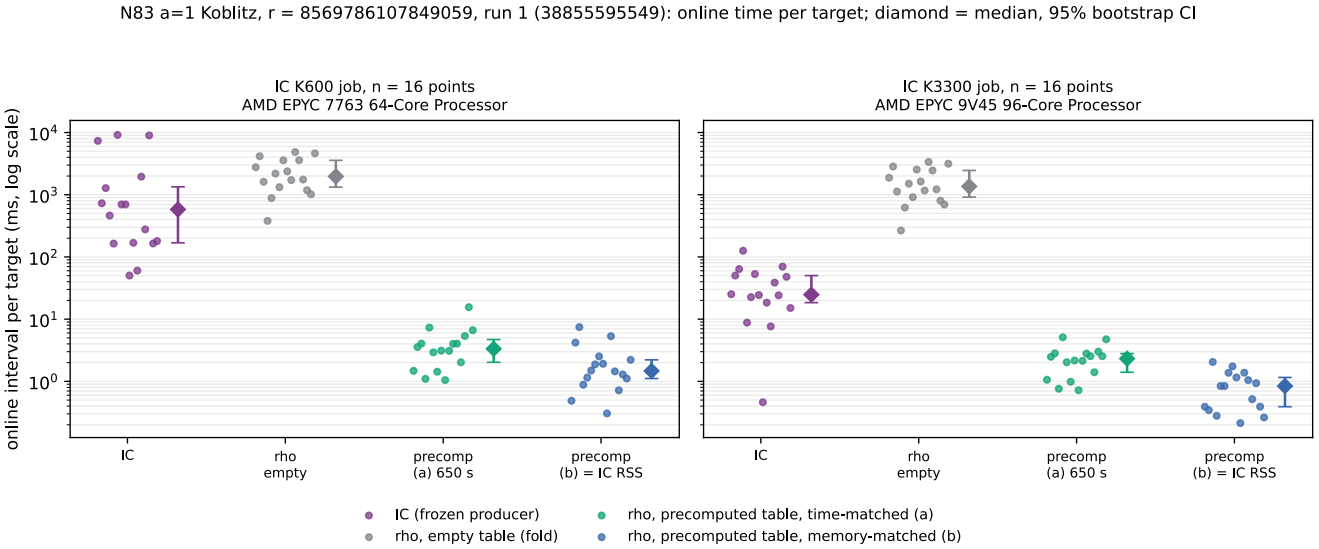


Figure 2: Run 1 (38855595549), the protocol as first frozen. The k3300 job's arm (a) had more wall than IC's setup on that runner (amendment 1).

4. Method notes, deviations and limits

- **Producers.** IC: frozen `koblitz_orbit_dlp_fast.rs` @ `d0e4aa13` (K3300, Tier 1 root11 environment, rank seed 29, 4 rank threads) and `koblitz_orbit_dlp_fast_online.rs` @ `eaa75494` (K600, rank seed 7, one CPU), each built from an archive of its commit. Empty-table rho: frozen `n83_rho_optimization_eval.rs` @ `d0e4aa13` with `rho_config.json` (fold, 64 lanes, dp 6). Precomputed rho: `examples/n83_precomputed_rho.rs` on `StrongRhoG::dp_walks`, same fold field, jumps, partition, DP hash and canonicalization as `solve`.
- **Deviation 1 (declared).** One IC process per job receives all 16 targets (setup once, per-target online clock); the original runs started a process per target. The online interval's code path is unchanged.
- **Deviation 2 (amendment 1).** Runner classes vary on hosted `ubuntu-24.04`: run 1 k3300 EPYC 9V45, run 1 k600 and run 2 k3300 EPYC 7763, run 2 k600 Intel Xeon 6973P-C. Every ratio is within one job on one runner. Arm (a′) was added before run 2 so the time budget follows the IC's measured setup wall on the runner it lands on.
- **Seed variance.** Arms (a) and (a′) differ in table/jump seed. Arm (a)'s per-target step counts are identical across all jobs (deterministic walks from the same seed), so each is one draw of the walk randomness; (a′)'s median online steps were about 2× (a)'s. Both draws give IC/rho intervals above 1.
- **Online interval of the precomputed arms** includes an on-curve and subgroup check of the target (about 0.03 ms) that IC's interval does not include, and the fast `[d]G = Q` recovery check. The generic replay runs outside it.
- **Precomputation accounting.** (a′) uses one thread for a wall equal to IC's target-independent wall; IC K3300 used 4 rank threads (825 s user CPU on 682 s wall in run 2), so (a′) had less CPU than IC. (b) used 4 threads until its table reached the IC's maximum RSS (1,904–5,293 s wall): memory-matched, not time-matched.

- **Isolation.** Hosted shared runners, no `isolated_bench.py` reservation (L0 / cloud-shared). Host manifests (`lscpu`, `meminfo`, cgroup) are in each job's `host/`.
- **CI tests.** Run 1: library 4,146 passed, 0 failed; example tests pass. Run 2′s first attempt failed two `jv_cover` tests on wall-clock timeouts (`timed_out: true`) in code this PR does not touch; logs in `records/run-38878565324/tests/`.
- **Scope.** Public synthetic hash-to-curve points on one curve; online cost at matched reusable precomputation, not cold cost and not an asymptotic statement. No key recovery or deployed-curve claim.
- **Prediction check.** `PROTOCOL.md` predicted ≈ 5–15 ms online for (a) and $\lesssim$ 1 ms for (b). Measured medians: (a)/(a′) 2.3–8.1 ms, (b) 0.84–1.47 ms.